

Remark D.21. We remark that $u = \arg \max_{v \in S_i \cup \{i\}} f_{S_i, i}(S_i \cup \{i\} \setminus v) - f_{S_i, i}(S_i)$ is the same as $u = \arg \max_{v \in S_i \cup \{i\}} f_{S_i, i}(S_i \cup \{i\} \setminus v)$. For the convenience of discussion and analysis, we will switch to using both cases in different places.

Here, we explain how to add items into sets via approximate values.

Definition D.22 (Expanding items based on approximate value). *If the following conditions hold*

- Let $S_i \subseteq [i - 1]$.
- Let $\tilde{f}_{S_i, i} : 2^{[i]} \rightarrow \mathbb{R}$.

Then, we can define S_{i+1} as follows

- If $|S_i| = k$ then $S_{i+1} = S_i \cup \{i\} \setminus u$ where $u = \arg \max_{v \in S_i \cup \{i\}} \tilde{f}_{S_i, i}(S_i \cup \{i\} \setminus v) - \tilde{f}_{S_i, i}(S_i)$
- If $|S_i| < k$, then $S_{i+1} = S_i \cup \{i\}$.

D.11 Universal Conditions

We state several definitions for both the exact function and approximation function.

Definition D.23 (Universal Monotone Condition (for exact function)). *We say f has universal monotone condition, if for all $i \in [n]$ for all $S_i \subseteq [i - 1]$, we have*

$$f_{S_i, i}(X) \geq f_{S_i, i}(Y), \quad \forall Y \subset X$$

Definition D.24 (Universal Monotone Condition (for approximate function)). *We say $\tilde{f}$ has universal monotone condition, if for all $i \in [n]$ for all $S_i \subseteq [i - 1]$, we have*

$$\tilde{f}_{S_i, i}(X) \geq \tilde{f}_{S_i, i}(Y), \quad \forall Y \subset X$$

Definition D.25 (Universal Dynamic Condition 1 (for exact function)). *We say f has universal dynamic condition 1 (for exact function), if for all $i \in [n]$ for all $S_i \subseteq [i - 1]$, $S_{i-1} \subseteq [i - 2]$, $|S_i \setminus S_{i-1}| \leq 1$, we have*

$$f_{S_i, i}(S_i) \geq (1 - \theta) \cdot f_{S_{i-1}, i-1}(S_i)$$

Definition D.26 (Universal Dynamic Condition 1 (for approximate function)). *We say $\tilde{f}$ has universal dynamic condition 1 (for approximate function), if for all $i \in [n]$ for all $S_i \subseteq [i - 1]$, $S_{i-1} \subseteq [i - 2]$, $|S_i \setminus S_{i-1}| \leq 1$, we have*

$$\tilde{f}_{S_i, i}(S_i) \geq (1 - \theta) \cdot \tilde{f}_{S_{i-1}, i-1}(S_i)$$

We define opt_i as follows:

Definition D.27. *Let k denote the budget length. For each $i \in [n]$, we define opt_i as*

$$\max\{f_{X, i}(Y) \mid X \subseteq [i - 1], Y \subseteq [i], |X| \leq k, |Y| \leq k, |Y \setminus X| \leq 1\}.$$

Definition D.28 (Universal Dynamic Condition 2). *For $i \in [n]$, we define opt_i as Definition D.27. We say it has universal dynamic condition if for each $i \in [n]$, we have*

$$\text{opt}_i \geq (1 - \gamma) \cdot \text{opt}_{i-1}.$$

D.12 Induction Lemma for Exact Function

The goal of this section is to prove Lemma D.29

Lemma D.29 (Induction Lemma). *For a fixed i , suppose the following conditions hold*

- **Set Condition.** $S_i \subseteq [i - 1]$
- **Budget Condition.** $|S_i| \leq k$

- **Value Condition.** $f_{S_{i-1},i-1}(S_i) \geq (1 - 1/e) \cdot (1 - \theta)^i (1 - \gamma)^i \text{opt}_i$
- **Universal Dynamic Condition 1 (for exact function).** (See Definition D.25) $f_{S_i,i}(S_i) \geq (1 - \theta) \cdot f_{S_{i-1},i-1}(S_i)$
- **Universal Dynamic Condition 2.** (See Definition D.28) $\text{opt}_i \geq (1 - \gamma) \cdot \text{opt}_{i+1}$
- **Universal Monotone Condition (for exact function).** (See Definition D.23) $f_{S_i,i}(X) \geq f_{S_i,i}(Y)$ for all $Y \subset X$

Then if we construct S_{i+1} as Definition D.20, then we have

- **Set Condition.** $S_{i+1} \subseteq [i]$
- **Budget Condition.** $|S_{i+1}| \leq k$
- **Value Condition.** $f_{S_i,i}(S_{i+1}) \geq (1 - 1/e) \cdot (1 - \theta)^{i+1} (1 - \gamma)^{i+1} \text{opt}_{i+1}$

Proof. **Proof of Set Condition.**

Note that $S_i \subseteq [i - 1]$, by using the way we construct S_{i+1} , then it is obvious that $S_{i+1} \subseteq [i]$.

Proof of Budget Condition.

Note that $|S_i| \leq k$, by using the way we construct S_{i+1} , then it is straightforward that $|S_{i+1}| \leq k$.

Proof of Value Condition.

We can show that

$$\begin{aligned}
 f_{S_i,i}(S_{i+1}) &\geq f_{S_i,i}(S_i) \\
 &\geq (1 - \theta) \cdot f_{S_{i-1},i-1}(S_i) \\
 &\geq (1 - \theta) \cdot ((1 - 1/e)(1 - \theta)^i (1 - \gamma)^i \text{opt}_i - i \cdot \epsilon_0) \\
 &\geq (1 - \theta) \cdot ((1 - 1/e)(1 - \theta)^i (1 - \gamma)^{i+1} \text{opt}_{i+1}) \\
 &\geq (1 - 1/e) \cdot (1 - \theta)^{i+1} (1 - \gamma)^{i+1} \text{opt}_{i+1}.
 \end{aligned}$$

where the first step follows from **Universal Monotone Condition**, the second step follows from **Universal Dynamic Condition 1**, the third step follows from **Value Condition**, the fourth step follows from **Universal Dynamic Condition 2**, and the last step follows from simple algebra.

Thus, we complete the proof. $\square$

D.13 Induction Lemma for Approximate Function

The goal of this section is to prove Lemma D.30

Lemma D.30 (Induction Lemma). *For a fixed i , suppose the following conditions hold*

- **Set Condition.** $S_i \subseteq [i - 1]$
- **Budget Condition.** $|S_i| \leq k$
- **Value Condition.** $f_{S_{i-1},i-1}(S_i) \geq (1 - 1/e) \cdot (1 - \theta)^i (1 - \gamma)^i \text{opt}_i - i \cdot \epsilon_0$
- **Universal Dynamic Condition 1 (for approximate function).** (see Definition D.26) $\tilde{f}_{S_i,i}(S_i) \geq (1 - \theta) \cdot \tilde{f}_{S_{i-1},i-1}(S_i)$
- **Universal Dynamic Condition 2.** $\text{opt}_i \geq (1 - \gamma) \cdot \text{opt}_{i+1}$
- **Universal Approximate Condition.** (See Definition D.28) $f_{S_i,i}(X) \geq \tilde{f}_{S_i,i}(X) - \epsilon_0$ for all X
- **Universal Monotone Condition (for approximate function).** (see Definition D.24) $\tilde{f}_{S_i,i}(X) \geq \tilde{f}_{S_i,i}(Y)$ for all $Y \subset X$

Then if we construct S_{i+1} as Definition D.22, then we have

- **Set Condition.** $S_{i+1} \subseteq [i]$
- **Budget Condition.** $|S_{i+1}| \leq k$
- **Value Condition.** $f_{S_{i+1},i}(S_{i+1}) \geq (1 - 1/e) \cdot (1 - \theta)^{i+1} (1 - \gamma)^{i+1} \text{opt}_{i+1} - (i + 1) \cdot \epsilon_0$

Proof. **Proof of Set Condition.**

Note that $S_i \subseteq [i - 1]$, by using the way we construct S_{i+1} , then it is obvious that $S_{i+1} \subseteq [i]$.

Proof of Budget Condition.

Note that $|S_i| \leq k$, by using the way we construct S_{i+1} , then it is straightforward that $|S_{i+1}| \leq k$.

Proof of Value Condition.

We can show that

$$\begin{aligned}
f_{S_{i+1},i}(S_{i+1}) &\geq \tilde{f}_{S_{i+1},i}(S_{i+1}) - \epsilon_0 \\
&\geq \tilde{f}_{S_{i+1},i}(S_i) - \epsilon_0 \\
&\geq (1 - \theta) \cdot \tilde{f}_{S_{i-1},i-1}(S_i) - \epsilon_0 \\
&\geq (1 - \theta) \cdot ((1 - 1/e)(1 - \theta)^i (1 - \gamma)^i \text{opt}_i - i \cdot \epsilon_0) - \epsilon_0 \\
&\geq (1 - \theta) \cdot ((1 - 1/e)(1 - \theta)^i (1 - \gamma)^{i+1} \text{opt}_{i+1} - i \cdot \epsilon_0) - \epsilon_0 \\
&\geq (1 - 1/e) \cdot (1 - \theta)^{i+1} (1 - \gamma)^{i+1} \text{opt}_{i+1} - (i + 1) \cdot \epsilon_0.
\end{aligned}$$

where the first step follows from **Universal Approximate Condition**, the second step follows from **Universal Monotone Condition**, the third step follows from **Universal Dynamic Condition 1**, the forth step follows from **Value Condition**, the fifth step follows from **Universal Dynamic Condition 2**, and the last step follows from simple algebra.

Thus, we complete the proof. $\square$

D.14 Theoretical Result

We first give the guarantee of our full-knowledge version (without cache size limitation).

Lemma D.31 (Formal version of Lemma 3.1). *Under the mild assumption, let k denote any target size. If we greedily compute the attention score based on full information, then we can find the set S_i such that*

$$f(S_i) \geq (1 - 1/e) \cdot (1 - \alpha) \text{opt}_i,$$

where $\alpha \in (0, 1)$ are parameters.

Proof. The proof follows from using Theorem D.17, Corollary D.18, Lemma D.29 with choosing $\theta = \gamma = \alpha/(10n)$. $\square$

Next, we show the guarantee for our robust and approximate greedy eviction policy algorithm (Algorithm 2).

Theorem D.32 (Formal version of Theorem 4.4). *Under the mild assumption, let k denote the budget of space limitation. If for each token, we greedily compute the attention score based on top- k choice, then we can show the set $\tilde{S}_i$ we generate each for token $i \in [n]$ satisfy that*

$$f(\tilde{S}_i) \geq (1 - 1/e) \cdot (1 - \alpha) \text{opt}_i - \beta,$$

where $\alpha \in (0, 1), \beta > 0$ are parameters.

Proof. The proof follows from using Theorem D.17, Corollary D.18, Lemma D.30 with choosing $\epsilon_0 = \beta/(10n)$ and $\theta = \gamma = \alpha/(10n)$. $\square$